radius = $\frac{1}{2}$ x diameter

Name?

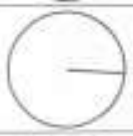
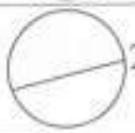
Area of Circles

Write the formla for the area of a circle in the box below.

$A = r^2 \times Pi$

$A = \left(\frac{1}{2}d\right)^2 \times pi$

Find the area of each circle to the nearest tenth. Show your work in the space provided.

Given Information	Work	Area
1. radius = 19 in	19x19=36.1 xPi=	113.354in
2. diameter = 3 ft	1.5² =2.25 xPi=	7.065 ft
3. diameter = 2.7 in	÷2=1.35²=1.8225 xπ=	5.72265 in
4. diameter = 3¼ m	3.25 ÷2 = 1.625 x2 =2.64125 x3.14	8.293525 m
5. radius = 28 cm	28x28 = 784 x 3.14=	2461.76cm
6. 12 cm	R 12x12=144 x Pi = 452.16	452.16cm
7. 2.5 m	R 2.5² =6.25 xPi = 19.625	19.625m
8.  2½ in	R 2.5² =6.25 xPi=	19.625in
9. 8 ft	D ÷2=4² =16 * Pi = 56.24	50.25 ft
10.  23 in	D ÷2=11.5² =132.25xPi=	415.265

Area and Circumference Word Problems

For each word problem state whether you need to find the **area** or **circumference** and circle the word or phrase that tells you which you are looking for. Then solve the problem. Round your answer to the nearest tenth.

1. You buy a circular rug with a diameter of 8 feet. How much floor space will the rug cover?

Area = 8 ÷ 2 = 4 x 4 = 16 x 3.14 = 50.24 ft²

2. Terry ran around a circular track. If the radius of the track is 200 feet, how far did Terry run?

C = 2 x 200 = 400 x Pi = 1256 ft

3. Sue had a circular picture that she wanted to frame. If the picture's diameter was 7.5 inches, how much framing material did she need?

C = 7.5 x Pi = 23.55 in

4. Tommy wanted to hang a circular poster on his wall. If the radius of the poster is 2 feet, how much space will the poster take up on the wall?

A = 2 x 2 = 4 x 3.14 = 12.56 ft²

5. Johnny wrapped a piece of string around a can of soup. If the soup's diameter is 4 inches, how long was the string Johnny used to wrap around it?

C = Pi Pi x 4 12.56

12.56 in

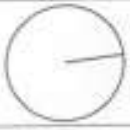
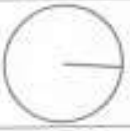
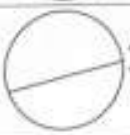
[redacted]

Area of Circles

Write the formula for the area of a circle in the box below.

$\pi = 3.14$ $A = r^2 \times \pi$ $A = (\frac{1}{2}D)^2 \times \pi$

Find the area of each circle to the nearest tenth. Show your work in the space provided.

Given Information	Work	Area
1. radius = 19 in	$19^2 \times 3.14 =$	1133
2. diameter = 3 ft	$1.5^2 \times 3.14$	7.065
3. diameter = 2.7 in	$1.35^2 \times 3.14$	5.72265
4. diameter = 3¼ m	A 1.625×3.14	8.2915625
5. radius = 28 cm	$A = 28 \times 3.14$	87.92
6. 12 cm	$A = 12^2 \times 3.14$	452.16 cm²
7.  2.5 m	$A = 2.5^2 m \times 3.14$	19.625 m²
8.  2½ in	$A = 2.5^2 in \times 3.14$	19.625 in²
9. 8 ft D	$A = 4^2 \times 3.14$	50.24
10.  23 in D	$A = 11.5^2 \times 3.14$	415.265

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3. Sue had a circular picture that she wanted to frame. If the picture's diameter was 7.5 inches, how much framing material did she need?

   C=7.5×3.14=23.55 in

4. Tommy wanted to hang a circular poster on his wall. If the radius of the poster is 2 feet, how much space will the poster take up on the wall?

   A=2×2=4×3.14=12.56 ft²

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   C=Pi×4=12.56 in

Area and Circumference Word Problems

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$C = Pi \times 4 = 12.56 in$